ExecuAI – My Contribution Document

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My Responsibilities

Enterprise Knowledge Base (RAG) Development

- Semantic Search Engine
- Document Processing & Embedding Pipeline
- Vector Database Design (ChromaDB)
- Document Version Comparison

Enterprise Knowledge Base (RAG)

Designed the Enterprise Knowledge Base that enables the AI Assistant to retrieve accurate information from company documents. Responsible for document ingestion, text extraction, chunking, embedding generation, semantic retrieval, and providing relevant context to the AI Assistant.

Workflow:

Company Documents → Upload → Text Extraction → Chunking → Embeddings → ChromaDB → Semantic Search → AI Assistant

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Semantic Search Engine

Designed a semantic search system that retrieves relevant document content based on the meaning of a user's query rather than exact keyword matching. This allows employees to quickly find policies, reports, meeting notes, and enterprise knowledge.

Example:

User Query → Generate Embedding → Similarity Search → Top Relevant Chunks → Return Context

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Enterprise Document Processing

Designed the complete document processing pipeline for enterprise documents.

Supports:

- PDF Documents
- DOCX Documents
- TXT Files

Features:

- Text Extraction

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- Intelligent Chunking
- Metadata Generation
- Embedding Creation
- Document Indexing

Vector Database (ChromaDB)

Designed the vector database structure to store document embeddings and metadata for fast semantic retrieval.

Metadata includes:

- Document Name
- Page Number
- Chunk ID
- Department
- Version
- Upload Date

Planned integration with FastAPI APIs.

Document Version Comparison

Designed a document comparison module to identify changes between different versions of enterprise documents.

Features:

- Added Content Detection
- Deleted Content Detection
- Modified Section Identification
- Version History Tracking

Current Progress

Completed:

- RAG Architecture Design
- Document Processing Workflow
- ChromaDB Planning
- API Structure Design

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- Start
- │
- ▼
- Install Cursor IDE
- │
- ▼
- Login to Cursor
- │
- ▼
- Create Project Folder (ExecuAI)
- │
- ▼
- Open ExecuAI Folder in Cursor
- │
- ▼
- Verify Python Installation
- │
- ▼
- Create Virtual Environment (venv)
- │
- ▼
- Activate Virtual Environment
- │
- ▼
- Install Required Packages
- (FastAPI, Uvicorn, python-dotenv)
- │
- ▼
- Create Backend Folder Structure
- │
- ▼
- Create main.py
- │
- ▼
- Run FastAPI Server
- │
- ▼
- Verify API in Browser
- (http://127.0.0.1:8000)
- │
- ▼
- Project Setup Completed ✅
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In Progress:

- Document Upload APIs
- Embedding Pipeline
- Semantic Search APIs
- Version Comparison Module

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Flowchart: Enterprise Documents

(PDF / DOCX / TXT Files)

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Document Upload API

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Text Extraction

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Chunking (500 + Overlap)

▼

Generate Text Embeddings

▼

Store Vectors in ChromaDB

▼

Semantic Search Engine

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Top Relevant Document Chunks

▼

AI Assistant (ExecuAI Response) hello

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